

The Gram double pencil: a spectral census for the zeros of ζ , an Euler anchor, and a bank-generated warp (Version 2)

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Abstract

We attach to each ordinate window a pair of Hankel matrices computed by contour integration of ξ'/ξ , and prove that the number of zeros off the critical line in the window equals the negative inertia of the first matrix (Theorem 2); the remaining spectral data of the pair counts the zeros with multiplicity, their distinct support, and their ordinates (Theorem 5). Both statements are unconditional. On symmetric pairs of windows, the location statement is equivalent to a Stieltjes-type positivity condition in the quotient chart $w = z^2$ (Proposition 6). We then construct, in the half plane of absolute convergence and with no reference to zeros, an Euler-anchored positive-definite double pencil (Proposition 8), an explicit finite-dimensional regulator (Theorem 9), and an exact bank-generated warp for every regular Hermitian-definite pencil path (Theorems 14 and 15). These finite-dimensional results do not identify the terminal pencil with the contour-moment pencil. That endpoint identification is isolated as the seating hypothesis in §8; the analytic defect calculation needed to test it is stated in §9.

1 Setup

Let ξ be the completed Riemann zeta function and put

$$A(z) := \xi\left(\frac{1}{2} + iz\right),$$

an entire function of order one, even ($A(-z) = A(z)$, the functional equation), real on $\mathbb{R}$, whose zeros are

$$z_\rho = -i\left(\rho - \frac{1}{2}\right) = \gamma_\rho + i\left(\frac{1}{2} - \beta_\rho\right)$$

for $\rho = \beta_\rho + i\gamma_\rho$ a nontrivial zero of ζ ; in particular

$$\rho \text{ on the critical line} \iff z_\rho \in \mathbb{R}.$$

Nonreal zeros of A occur in conjugate pairs $z, \bar{z}$ with equal multiplicities (conjugate symmetry of ξ on the real axis).

Definition 1 (window moments). Let $W = (a, b)$ with A nonvanishing on $\partial W \times \{|y| \leq Y\}$, $Y > \frac{1}{2}$, and let ∂R_W be the positively oriented rectangle boundary. Put

$$\mu_k(W) := \frac{1}{2\pi i} \oint_{\partial R_W} z^k \frac{A'(z)}{A(z)} dz, \quad k \geq 0,$$

and for $n \geq 1$,

$$H_n(W) := (\mu_{\alpha+\beta}(W))_{\alpha,\beta=0}^{n-1}, \quad H_n^{(1)}(W) := (\mu_{\alpha+\beta+1}(W))_{\alpha,\beta=0}^{n-1}.$$

By the argument principle, with $x_1, \dots, x_r$ the distinct real zeros of A in the window (multiplicities $a_i \geq 1$) and $z_j, \bar{z}_j, j = 1, \dots, q$, the distinct conjugate pairs (equal multiplicities $b_j \geq 1$),

$$\mu_k(W) = \sum_{i=1}^r a_i x_i^k + \sum_{j=1}^q b_j (z_j^k + \bar{z}_j^k). \quad (1)$$

All $\mu_k(W)$ are real and computable from A alone; no knowledge of the zeros is used. Write $m := r + 2q$ for the number of distinct support points.

2 The inertia identity

Theorem 2 (inertia). *For every $n \geq m$,*

$$\text{inertia } H_n(W) = (r + q, q, n - r - 2q),$$

where inertia denotes (positive, negative, zero) eigenvalue counts. In particular

$$n_-(H_n(W)) = q, \quad \text{and} \quad H_n(W) \succeq 0 \iff q = 0,$$

i.e. the negative inertia counts exactly the off-critical-line conjugate pairs.

Proof. Let $v(t) := (1, t, \dots, t^{n-1})^T$. By (1),

$$H_n = \sum_{i=1}^r a_i v(x_i) v(x_i)^T + \sum_{j=1}^q b_j \left[v(z_j) v(z_j)^T + v(\bar{z}_j) v(\bar{z}_j)^T \right].$$

Write $v(z_j) = A_j + iB_j$ with $A_j, B_j \in \mathbb{R}^n$; then $v(\bar{z}_j) = A_j - iB_j$ and

$$v(z_j) v(z_j)^T + v(\bar{z}_j) v(\bar{z}_j)^T = 2A_j A_j^T - 2B_j B_j^T, \quad (2)$$

the minus sign being the whole mechanism. Hence $H_n = UJU^T$ with

$$U = [v(x_1) \cdots v(x_r) \ A_1 \ B_1 \cdots A_q \ B_q], \quad J = \text{diag}(a_1, \dots, a_r, 2b_1, -2b_1, \dots, 2b_q, -2b_q),$$

so inertia $J = (r + q, q, 0)$.

U has full column rank m : the complex matrix $V = [v(x_1), \dots, v(x_r), v(z_1), v(\bar{z}_1), \dots]$ is Vandermonde on m distinct nodes with $n \geq m$, so its leading $m \times m$ minor equals $\prod_{\alpha < \beta} (s_\beta - s_\alpha) \neq 0$ and rank $V = m$; replacing each pair $(v(z_j), v(\bar{z}_j))$ by $(A_j, B_j) = (\frac{v(z_j) + v(\bar{z}_j)}{2}, \frac{v(z_j) - v(\bar{z}_j)}{2i})$ is an invertible change of columns, so rank $U = m$.

Take a thin QR factorisation $U = QR$ with $Q^T Q = I_m$ and R invertible, and extend Q to an orthogonal $\tilde{Q} = [Q, Q_\perp]$. Then

$$\tilde{Q}^T H_n \tilde{Q} = \begin{pmatrix} RJR^T & 0 \\ 0 & 0_{n-m} \end{pmatrix},$$

and by Sylvester's law of inertia $\text{inertia}(RJR^T) = \text{inertia}(J)$. The zero block contributes $n - m$ null directions. $\square$

Remark 3. The hypothesis $n \geq m$ is necessary: an off-line pair is invisible to a Hankel matrix smaller than the distinct support count. Since $\mu_0(W)$ is the number of zeros in W with multiplicity and $m \leq \mu_0(W)$, the choice $n := \max\{1, \mu_0(W)\}$ is admissible and computable from the contour data alone.

Remark 4 (collisions). If a conjugate pair approaches $\mathbb{R}$, the corresponding negative eigenvalue tends to 0^- and at collision becomes a null direction; the invariant through such an event is the inertia, not any individually ordered eigenvalue. Consequently the correct target is non-negativity $H_n \succeq 0$ (equivalently $n_- = 0$), never strict positivity: the $n - m$ structural nulls and the collision nulls are expected.

3 The census, and the equivalent form of RH

Theorem 5 (spectral census). *Let $n \geq m$. From the spectral data of $H_n(W)$ and the value $\mu_0(W)$:*

$$\mu_0 = \text{number of zeros in } W \text{ with multiplicity}, \quad \text{rank } H_n = m = r + 2q,$$

$$n_-(H_n) = q, \quad n_+(H_n) - n_-(H_n) = r,$$

and $\mu_0 > \sum_i 1 + \sum_j 2$ (i.e. μ_0 exceeds the distinct count) if and only if some zero in W is multiple. Moreover, if $Q \in \mathbb{R}^{n \times m}$ has orthonormal columns spanning $\text{ran } H_n$, then the generalized eigenvalues of the compressed pencil

$$\hat{H}_1 := Q^T H_n^{(1)} Q, \quad \hat{H}_0 := Q^T H_n Q$$

are exactly the support points x_i and $z_j, \bar{z}_j$.

Proof. The first four items are Theorem 2 together with $\mu_0 = \sum a_i + 2 \sum b_j$. For the pencil statement, enumerate the complex support as $s_1, \dots, s_m$ and its positive multiplicities as $d_1, \dots, d_m$. Put

$$V = [v(s_1) \cdots v(s_m)], \quad D = \text{diag}(d_1, \dots, d_m), \quad \Sigma = \text{diag}(s_1, \dots, s_m).$$

Then, with ordinary transpose (not conjugate transpose),

$$H_n = V D V^T, \quad H_n^{(1)} = V D \Sigma V^T.$$

The Vandermonde argument in Theorem 2 gives $\text{rank } V = m$ and $\text{ran}_{\mathbb{C}} H_n = \text{ran}_{\mathbb{C}} V$. Thus $B := Q^T V$ is invertible and

$$\det(\hat{H}_1 - \lambda \hat{H}_0) = \det(B)^2 \det(D) \prod_{a=1}^m (s_a - \lambda).$$

The compressed determinant is not identically zero and its roots, with multiplicity, are precisely the support points. Conjugate support pairs make the determinant real. $\square$

Proposition 6 (symmetric quotient chart; Stieltjes form). *Let $0 \leq a < b$ and let $W^\pm$ denote the union of the two reflected ordinate windows (a, b) and $(-b, -a)$, with boundary contours chosen away from zeros. Define $\mu_k(W^\pm)$ by adding the two contour integrals and set $\nu_k(W^\pm) := \mu_{2k}(W^\pm)$. Then the zero multiset in $W^\pm$ is invariant under $z \mapsto -z$, and ν_k is the moment sequence of the pushforward counting measure under $w = z^2$. Under this map a real zero maps*

to $w \geq 0$, while every nonreal orbit produces either a nonreal conjugate pair in the w -plane or a negative real support point. Consequently, for n at least the number of distinct w -support points,

$$\text{all zeros of } A \text{ in } W^\pm \text{ are real} \iff (\nu_{\alpha+\beta}) \succeq 0 \text{ and } (\nu_{\alpha+\beta+1}) \succeq 0,$$

the finite Stieltjes condition for the pushforward measure. Hence the zero-location statement is equivalent to positivity of both even-moment Hankel matrices on every member of a symmetric paired tiling.

Proof. The use of $W^\pm$, rather than a single generic interval, ensures that restriction to the window commutes with the even quotient. Pairing the terms at z and $-z$ gives a positive integer weight at $w = z^2$. For a finite real moment sequence, positivity of the Hankel matrix and its shift is equivalent to support in $[0, \infty)$; the Vandermonde inertia argument of Theorem 2 supplies the converse directly at any order at least the distinct support count. Finally $z^2 \in [0, \infty)$ holds exactly when $z \in \mathbb{R}$. $\square$

Corollary 7 (sign rigidity). *Let $t \mapsto A_t$ be a continuous deformation with zeros varying continuously and no zero crossing ∂W . At every parameter for which n dominates the distinct support count,*

$$q(t) = n_-(H_n(t)).$$

Thus the conjugate-pair count changes exactly when the negative inertia changes, and such a change can occur only at a zero eigenvalue of $H_n(t)$. A tangency at zero need not change the count. In particular, if $H_n(t) \succeq 0$ for the entire deformation, then $q(t) = 0$ throughout. Positivity at a single parameter does not by itself propagate; propagation requires exclusion of a subsequent negative-inertia change.

4 Unconditional statements

The following hold with no hypothesis.

- (U1) For every window W and $n \geq m(W)$: $q(W) = n_-(H_n(W))$; the off-critical-line pair count of a window is the negative inertia of a matrix computed by contour integration of A'/A .
- (U2) The census of Theorem 5 holds as stated.
- (U3) The zero-location statement is equivalent to the Stieltjes condition of Proposition 6 on every member of a symmetric paired tiling.
- (U4) For every window contained in the numerically verified range ($|t| \leq 3 \cdot 10^{12}$ by the rigorous interval computation of Platt and Trudgian [1]), $q(W) = 0$ and hence $H_n(W) \succeq 0$: the positivity hypothesis is an established fact on an initial segment of the tiling.
- (U5) With $Y > \frac{1}{2}$ and the convention $A(z) = \xi(\frac{1}{2} + iz)$, the bottom edge of ∂R_W lies in the half plane of absolute convergence and the top edge maps there by the functional equation, giving a decomposition $H_n = H^{\text{pr}} + H^{\text{arch}} + H^{\text{side}}$ into blocks computable from Dirichlet data, Γ -factors and the lateral edges respectively; in a tiling, adjacent lateral contributions cancel pairwise.

5 The Euler-anchored pencil and its regulator

Nothing in this section refers to zeros; the dependency order is

Euler/FE bank $\rightarrow (G_1, G_0) \rightarrow$ real spectrum $\rightarrow$ contour transport $\rightarrow$ residues $\rightarrow$ zeros,
and no step may be inverted.

Proposition 8 (Euler anchor). *Fix $s_0 > 1$ and $N \geq 1$, and set*

$$G_\ell[j, k] := \sum_{n \geq 2} \Lambda(n) n^{-s_0} (\log n)^{j+k+\ell}, \quad 0 \leq j, k < N, \quad \ell \in \{0, 1\}.$$

The series converge absolutely, and G_0, G_1 are the Hankel matrices of the positive measure $\sum_n \Lambda(n) n^{-s_0} \delta_{\log n}$ supported on infinitely many points of $[\log 2, \infty)$. Hence

$$G_0 > 0, \quad G_1 > 0,$$

and $\text{spec}(G_1, G_0) \subset [\log 2, \infty)$ consists of the Gauss quadrature nodes of that measure.

Proof. Absolute convergence for $s_0 > 1$ is standard. Positivity is the Hamburger/Stieltjes criterion for a positive measure on $[\log 2, \infty)$: for $c \in \mathbb{R}^N$ and $P(x) = \sum_j c_j x^j$, $c^T G_0 c = \int P^2 d\sigma \geq 0$ and $c^T G_1 c = \int x P^2 d\sigma \geq 0$. If $c \neq 0$, the polynomial P has only finitely many roots, whereas the prime-power logarithms in the support are infinite; both integrals are therefore strictly positive. $\square$

Theorem 9 (finite-dimensional reverb regulator). *Let $t \mapsto (G_0(t), G_1(t))$ be a smooth family with $G_0(t) > 0$ on the active quotient, and set $S(t) := G_0^{-1/2} G_1 G_0^{-1/2}$, so $S = S^*$ and $\text{spec } S = \text{spec}(G_1, G_0)$. Suppose the active dimension is d , the spectrum of $S(t_0)$ is simple, with spectral projections P_a and unit eigenvectors v_a ($1 \leq a \leq d$), and the transport has exactly d scalar control channels,*

$$\dot{S} = F(u) = F_0 + \sum_{r=1}^d u_r F_r, \quad F_r = F_r^*.$$

Define the square response matrix $M \in \mathbb{R}^{d \times d}$ by $M_{ar} := \langle v_a, F_r v_a \rangle$ and put $b_a := -\langle v_a, F_0 v_a \rangle$. If $\det M \neq 0$, then $u := M^{-1}b$ is the unique control with $\Delta(F(u)) := \sum_a P_a F(u) P_a = 0$, and with

$$K := \sum_{a \neq b} \frac{P_a F(u) P_b}{\lambda_a - \lambda_b}, \quad K^* = -K,$$

one has $\dot{S} = [S, K]$. Consequently $S(t) = U(t)S(t_0)U(t)^$ with $\dot{U} = -KU$, U unitary, and*

$$\text{spec } S(t) = \text{spec } S(t_0) \quad \text{exactly.}$$

In particular the transported pencil remains Hermitian-definite and its generalized spectrum remains real.

Proof. First-order perturbation theory gives $\dot{\lambda}_a = \langle v_a, F v_a \rangle$, so $\Delta(F)$ is precisely the component of the flow able to move eigenvalues; the linear system $Mu = b$ expresses $\Delta(F(u)) = 0$ and is uniquely solvable iff $\det M \neq 0$. For such u , using $\sum_a P_a = I$,

$$[S, K] = \sum_{a \neq b} (\lambda_a - \lambda_b) \frac{P_a F(u) P_b}{\lambda_a - \lambda_b} = \sum_{a \neq b} P_a F(u) P_b = F(u) - \Delta(F(u)) = F(u) = \dot{S}.$$

$K^* = -K$ since $F(u)^* = F(u)$ and the denominators are real and antisymmetric in (a, b) . The solution of $\dot{U} = -KU$, $U(t_0) = I$, is unitary, and $\frac{d}{dt}(U^* S U) = U^* (\dot{S} - [S, K]) U = 0$. $\square$

Remark 10 (numerical verification). For the Euler anchor of Proposition 8 with $s_0 = 1.5$, $N = 4$, control channels given by reweighting the prime channels $p \in \{2, 3, 5, 7\}$ and raw flow $F_0 = \partial_{s_0} S$: the spectrum is simple $(0.9619, 2.7211, 5.6399, 8.8009)$, $\det M = -0.128$ with $\text{rank } M = 4$ and condition number $3.4 \cdot 10^2$; the solved control annihilates all diagonal drifts to 10^{-12} , and $\|[S, K] - (F - \Delta(F))\| = 1.3 \cdot 10^{-11}$ with $\|K + K^T\| = 1.7 \cdot 10^{-13}$.

Lemma 11 (block regulator at a degenerate spectrum). *Let $S = S^*$ have distinct eigenvalues Λ_α with spectral projections P_α of arbitrary finite multiplicities, and let the flow be controlled as in Theorem 9, $\dot{S} = F(u) = F_0 + \sum_r u_r F_r$, $F_r = F_r^*$. With*

$$\Delta(F) := \sum_{\alpha} P_{\alpha} F P_{\alpha}, \quad K := \sum_{\alpha \neq \beta} \frac{P_{\alpha} F P_{\beta}}{\Lambda_{\alpha} - \Lambda_{\beta}},$$

one has $[S, K] = F - \Delta(F)$ and $K^ = -K$. To first order the group Λ_α splits by the spectrum of the compression $P_\alpha F P_\alpha$; hence the regulated condition is the block equation $P_\alpha F(u) P_\alpha = 0$ for all α , a linear system in u generalizing $Mu = b$. Whenever that block system is solvable on an interval of fixed active dimension, the flow $\dot{S} = [S, K]$ is isospectral with multiplicities.*

Proof. $SP_\alpha = \Lambda_\alpha P_\alpha$ and $\sum_\alpha P_\alpha = I$ give

$$[S, K] = \sum_{\alpha \neq \beta} (\Lambda_\alpha - \Lambda_\beta) \frac{P_\alpha F P_\beta}{\Lambda_\alpha - \Lambda_\beta} = F - \Delta(F),$$

and $K^* = -K$ since $F^* = F$ and the real denominators are antisymmetric in (α, β) . The first-order motion of a degenerate group is degenerate perturbation theory: the group splits by the eigenvalues of $P_\alpha F P_\alpha$, which vanish identically iff the block equation holds; the condition is linear in u , and on solution intervals the Lax argument of Theorem 9 is unchanged. $\square$

Remark 12 (scope at singular events). Lemma 11 handles eigenvalue multiplicity for a fixed-dimensional Hermitian matrix. It does not by itself cross a rank drop of G_0 , prove solvability of the block response system, or extend the scalar interpolation warp through merging nodes. Those three assertions require separate continuation data and are retained explicitly in the terminal seating statement below.

6 Warp covariance: the bank-generated readout

Theorem 9 fixes the abstract spectrum $\{\lambda_a\}$ of the regulated flow; the same transport, unregulated, moves it. The first chart makes reality manifest, the second carries the physical ordinates,

and the map between them — the warp — is not a hypothesis to be assumed but an object the bank constructs. Throughout this section $t \mapsto (G_0(t), G_1(t))$ is a given smooth transport of the continued jet pair, whose inputs are values and derivatives of the continued logarithmic derivative along the path — function data; no zero location enters — and the parameter interval is *regular*: $G_0(t) > 0$ on the active quotient and $\text{spec } S_t$ simple, where $S_t := G_0^{-1/2} G_1 G_0^{-1/2}$. Theorems 14 and 15 apply on such regular intervals. Their hypotheses do not assert existence of the full path, positivity of G_0 along it, or gluing through collision and rank-drop events.

Definition 13 (bank velocity field; warp). Let $\lambda_1(t) < \dots < \lambda_n(t)$ be the eigenvalues of S_t , realized as the generalized eigenpairs $G_1 w_a = \lambda_a G_0 w_a$ normalized by $w_a^T G_0 w_a = 1$, and set

$$d_a(t) := w_a^T (\dot{G}_1 - \lambda_a \dot{G}_0) w_a \quad (= \langle v_a, \dot{S}_t v_a \rangle, \quad v_a = G_0^{1/2} w_a).$$

Let V_t be the unique real polynomial of degree $\leq n-1$ with $V_t(\lambda_a(t)) = d_a(t)$, and let the *warp* φ_t be the flow of the scalar equation $\dot{x} = V_t(x)$, $\varphi_0 = \text{id}$.

Theorem 14 (warp existence and exact covariance). *On every regular interval the warp is defined on a neighborhood of $[\lambda_1(0), \lambda_n(0)]$ and satisfies:*

- (i) **bank determination:** V_t , hence φ_t , is computed from $(G_0, G_1)(t)$ and $(\dot{G}_0, \dot{G}_1)(t)$ alone;
- (ii) **reality and order:** $\varphi_t(\mathbb{R} \cap \text{dom}) \subseteq \mathbb{R}$ and $x \mapsto \varphi_t(x)$ is strictly increasing, in particular injective;
- (iii) **exact covariance:** $\varphi_t(\lambda_a(0)) = \lambda_a(t)$ for every a and every t in the interval.

Proof. (i) is the construction. For (ii): V_t has real coefficients — real nodes, real diagonal drifts of a Hermitian family — so the flow preserves $\mathbb{R}$; the field is polynomial in x with coefficients continuous in t , hence locally Lipschitz in x uniformly on compact t -intervals, so distinct scalar characteristics never cross and φ_t is strictly increasing where defined. For (iii): first-order perturbation theory for the Hermitian-definite pencil gives $\dot{\lambda}_a = w_a^T (\dot{G}_1 - \lambda_a \dot{G}_0) w_a = d_a(t) = V_t(\lambda_a(t))$, so the eigenvalue path $t \mapsto \lambda_a(t)$ solves the initial-value problem defining $\varphi_t(\lambda_a(0))$, and uniqueness makes them equal. Existence: the node characteristics are the globally defined eigenvalue paths; a point between consecutive nodes is trapped between two of them by (ii); and continuous dependence on initial conditions extends the flow to a neighborhood of the compact hull for the same times. $\square$

Theorem 15 (determinant pullback). *Let $p_t(x) := \det(S_t - xI)$, and let I be an open real interval containing the time-0 active spectrum on which the warp is defined. Then on $\varphi_t(I)$,*

$$p_t = J_t \cdot (p_0 \circ \varphi_t^{-1}),$$

with J_t continuous and nonvanishing; consequently

$$Z(p_t) \cap \varphi_t(I) = \varphi_t(\text{spec } S_0) \subset \mathbb{R}.$$

Proof. The field is polynomial in x , so φ_t is a strictly increasing C^1 diffeomorphism of I onto $\varphi_t(I)$. On $\varphi_t(I)$ the functions p_t and $p_0 \circ \varphi_t^{-1}$ have the same zero set $\{\varphi_t(\lambda_a(0))\}_a = \{\lambda_a(t)\}_a$ (covariance and injectivity, Theorem 14), each zero simple (simple spectrum on a regular interval). The ratio $J_t := p_t / (p_0 \circ \varphi_t^{-1})$ is continuous and nonvanishing off the common zeros and extends across each with the nonzero limit $p'_t(\lambda_a(t)) / (p_0 \circ \varphi_t^{-1})'(\lambda_a(t))$. $\square$

Remark 16 (the two spectral charts). The regulated and unregulated flows describe one transport in two charts: labels frozen with a moving readout, or a moving spectrum read through the identity; by Theorem 14 their relation is φ_t . A constant-factor identity written with one spectral variable cannot express this change of chart unless the two root multisets already coincide. Moreover, when $n > m$, the full window determinant $\det(H_n^{(1)} - \lambda H_n)$ vanishes identically because of the structural nullspace. The correct finite determinant is the compressed determinant of Theorem 5; on every regular path its root motion is expressed by the pullback identity of Theorem 15.

Remark 17 (numerical verification). On the convergent segment ($s_0: 1.5 \rightarrow 1.2$, $N = 4$, exact jets of $-\zeta'/\zeta$; `tmp/att201.warp.probe.py`): the perturbation drift matches finite differences of the eigenvalue paths to $8.9 \cdot 10^{-8}$. The interpolated flow carries

$$(1.1916, 4.0395, 9.6214, 19.3384) \mapsto (2.1714, 9.2832, 23.2371, 47.5293)$$

with maximal error $1.9 \cdot 10^{-10}$; an interior test point remains strictly interior. The exact-jet anchor nodes differ from those quoted after Theorem 9, which realize the truncated prime sum at finite cutoff; the two banks are distinct legitimate realizations, and the drift between them is the truncation observation of the ledger.

7 The conditional theorem

Hypothesis 18 (PSD). For every window W of a tiling of $\mathbb{R}$ and $n = \max\{1, \mu_0(W)\}$, $H_n(W) \succeq 0$. Equivalently, on every symmetric paired window, both Hankel matrices of the even moments are positive semidefinite in the quotient chart of Proposition 6.

Theorem 19. *Assume Hypothesis 18. Then*

1. *every nontrivial zero of ζ lies on the critical line;*
2. *for each window the spectral data of $H_n(W)$ returns the full census of Theorem 5, with $n_- \equiv 0$;*
3. *the zeros are the real generalized eigenvalues of the compressed, explicitly computed symmetric pencils $(\hat{H}_1(W), \hat{H}_0(W))$ of Theorem 5.*

Proof. By Theorem 2, $H_n(W) \succeq 0$ with $n \geq m(W)$ forces $q(W) = 0$: no conjugate pair of zeros of A lies in W . Nonreal zeros occur in conjugate pairs with equal real part, hence both members lie in the same window of the tiling (for a pair on a boundary, apply the argument to a shifted tiling). Every nontrivial zero of ζ corresponds to a zero of A lying in some window, and only finitely many per window by Riemann–von Mangoldt. Hence all zeros of A are real, i.e. all nontrivial zeros of ζ have real part $\frac{1}{2}$. Items (2) and (3) are Theorem 5; under Hypothesis 18, $\hat{H}_0 > 0$ and $\hat{H}_1$ is real symmetric. $\square$

8 The seat: endpoint identification

Definition 20 (compressed terminal defect). Fix a window W , let $m = \text{rank } H_n(W)$, and choose $Q_W \in \mathbb{R}^{n \times m}$ with orthonormal columns spanning $\text{ran } H_n(W)$. Write

$$\hat{H}_0 := Q_W^T H_n(W) Q_W, \quad \hat{H}_1 := Q_W^T H_n^{(1)}(W) Q_W.$$

For a transported bank pencil whose terminal active dimension is m , identify its terminal active coordinates with those of Q_W and define

$$D_0 := G_0(1) - \hat{H}_0, \quad D_1 := G_1(1) - \hat{H}_1.$$

Both terms in each difference are obtained from function data: continued logarithmic- derivative jets on the bank side and contour integrals of A'/A on the window side. No full $n \times n$ determinant is used; when $n > m$ that determinant contains a common structural nullspace and is identically zero as a polynomial in the spectral parameter.

Hypothesis 21 (terminal seating (S)). For every window W there is a bank-generated transport from an Euler anchor to the window chart, decomposed into finitely many regular intervals, with the following properties:

1. on each regular interval $G_0 > 0$ on the active quotient and Theorem 14 supplies the real, order-preserving warp;
2. at every collision or rank change, the adjacent active spectra and warps admit compatible continuous multiset limits;
3. the terminal multiset of those limits equals $\text{spec}(\hat{H}_1, \hat{H}_0)$.

Every object used to construct the path and the transition data is required to come from Euler, functional-equation, jet, contour, and registration data, without inserting the support points as interpolation targets.

Theorem 22 (the conditional chain). *Hypothesis 21 implies Hypothesis 18, and therefore implies all three conclusions of Theorem 19.*

Proof. On each regular interval the generalized eigenvalues are real because $G_0 > 0$ and G_1 is real symmetric; equivalently, Theorem 14 carries the real anchor spectrum by a real map. The transition clause of Hypothesis 21 preserves reality under multiset limits. Its terminal clause therefore makes every root of the compressed determinant $\det(\hat{H}_1 - \lambda \hat{H}_0)$ real. By Theorem 5, these roots are exactly the zero support in W , so $q(W) = 0$. Theorem 2 now gives $H_n(W) \succeq 0$ for every window, which is Hypothesis 18. $\square$

Remark 23 (what the warp theorem proves). Theorems 14 and 15 prove an exact covariance statement for every specified regular Hermitian-definite path. They do not prove the existence of a path reaching the contour-moment pencil, continued positivity of G_0 , compatibility at singular events, or terminal equality with the compressed window spectrum. Those claims are precisely the content of Hypothesis 21; coordinate covariance alone is not endpoint identification.

9 Remaining analytic obligation

The endpoint comparison is to be performed on the compressed active pencils. Its computable entrywise form is the following defect evaluation:

$$G_\ell[j, k] = \frac{1}{2\pi i} \oint_{\Gamma} z^{j+k+\ell} \frac{A'(z)}{A(z)} dz + D_\ell[j, k], \quad \ell \in \{0, 1\},$$

where D_ℓ must be derived explicitly from the pole, Γ -factor, lateral-edge, windowing, and coordinate-compression terms. On $s_0 > 1$, the pole contribution has moments $k!/(s_0 - 1)^{k+1}$, the moments of $e^{-(s_0-1)x} dx$ on $[0, \infty)$, and is positive semidefinite. The signs and spectral effect of the remaining blocks must be retained rather than absorbed into a constant factor.

Once this identity and the event-gluing data establish Hypothesis 21, Theorem 22 applies. Until then, the unconditional results are the inertia identity, the compressed spectral census, Euler-anchor positivity, the regulator under its stated response hypotheses, and regular-interval warp covariance.

References

- [1] D. Platt and T. Trudgian, *The Riemann hypothesis is true up to $3 \cdot 10^{12}$* , Bulletin of the London Mathematical Society **53** (2021), 792–797, doi:10.1112/blms.12460.